

VEDIC TESTING LABORATORIES INDIA PVT. LTD.

Analytical & Food Testing Services Division | ISO/IEC 17025 Compliant Operational Standards
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LABORATORY QUALITY ANALYSIS REPORT

Report No:	VTL/2026/HC-SEP-REV	Report Date:	11-Sep-2026
Sample Received Date:	07-Sep-2026	Testing Period:	07-Sep-2026 to 10-Sep-2026
Customer Name:	HoneyChain Traceability Project (Demonstration Client)		
Customer Reference:	Auditor Verification Request - Ref # HC-AUD-2026-09		

SAMPLE INFORMATION

Sample Name:	Raw Honey	Batch Identification:	HC-SEP-REV
Sample Description:	Viscous liquid, amber color, typical natural honey odor	Packaging:	Sealed Glass Container (500g)
Sample Quantity:	2 x 500 g	Sample Condition:	Satisfactory, Intact Seal
Declared Origin:	Multifloral (Western Ghats Region)	Sampling Protocol:	Submitted by Client

PHYSICO-CHEMICAL & QUALITY PARAMETERS

S.N.	TEST PARAMETER	TEST METHOD / REFERENCE	UNIT	SPECIFIED LIMITS (FSSAI/ BIS)	OBSERVED VALUE	STATUS
1	Moisture Content	IS 4941:2010 / AOAC 969.38	% by mass	Max 20.0%	19.4%	PASS
2	Hydroxymethylfurfural (HMF)	IS 4941:2010 / AOAC 980.23	mg/kg	Max 80.0	24.5	PASS
3	Diastase Activity	IS 4941:2010 (Schade Scale)	DN unit	Min 8.0	14.2	PASS
4	Total Reducing Sugars	IS 4941:2010 / Lane & Eynon	% by mass	Min 65.0%	71.8%	PASS
5	Sucrose Content	IS 4941:2010 / AOAC 920.184	% by mass	Max 5.0%	2.1%	PASS
6	Fructose to Glucose Ratio (F/G)	IS 4941:2010 / HPLC	Ratio	Min 1.0	1.18	PASS
7	Free Acidity	IS 4941:2010 (Titrimetric)	meq/kg	Max 50.0	28.4	PASS
8	Electrical Conductivity	IHC Method 2009	mS/cm	Max 0.8	0.42	PASS
9	Water Insoluble Solids	IS 4941:2010 Gravimetric	% by mass	Max 0.1%	0.04%	PASS
10	Specific Gravity at 27°C	Pycnometer Method	-	Min 1.39	1.412	PASS

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ORGANOLEPTIC & SENSORY EXAMINATION

S.N.	PARAMETER	TEST REFERENCE	STANDARD REQUIREMENT	OBSERVED VALUE	STATUS
11	Appearance / Physical State	Sensory Evaluation	Viscous fluid, free from foreign matter	Conforms	PASS
12	Odour & Aroma	Sensory Evaluation	Characteristic pleasant floral aroma, free from foreign odours	Conforms	PASS
13	Taste / Flavour	Sensory Evaluation	Sweet, typical floral honey flavour, no fermentation/off-taste	Conforms	PASS
14	Color (Pfund Scale)	Optical Comparator	Amber to Dark Amber (Industry standard)	Amber (68 mm)	PASS

PURITY & ADULTERATION SCREENING

S.N.	TEST PARAMETER	TEST METHOD / REFERENCE	UNIT	ACCEPTABLE LIMIT	OBSERVED VALUE	STATUS
15	C4 Sugar Addition (EA-IRMS)	AOAC 998.12 / IRMS	%	Max 7.0%	Negative (< 1.5%)	PASS
16	C3 Sugar Screening (LC-IRMS)	Internal Method VTL-SOP-H04	-	Negative	Negative	PASS
17	Rice Syrup Marker (TMR)	LC-MS/MS Screening	-	Absent	Absent	PASS
18	Foreign Oligosaccharides	HPAEC-PAD	%	Absent	Not Detected	PASS
19	Specific Marker for Invert Sugar (SM-I)	HPLC-UV	mg/kg	Absent	Not Detected	PASS

Technical Notes & Analytical Observations:

- Moisture content measured at 19.4% is close to the upper statutory limit of 20.0% specified under FSSAI regulations. While fully compliant and within permissible parameters, proper sealed storage below 25°C is recommended to maintain stability.
- Diastase activity (14.2 DN) and HMF content (24.5 mg/kg) confirm that the sample has not been subjected to excessive heat treatment during processing or storage.
- Isotope Ratio Mass Spectrometry (EA-IRMS / LC-IRMS) and LC-MS/MS marker screens show no evidence of exogenous sugar addition (C3/C4 cane/corn syrups or commercial inverted rice syrups).

OVERALL LABORATORY CONCLUSION

SUMMARY STATEMENT: Laboratory analysis of sample Batch ID: **HC-SEP-REV** (Raw Honey) was conducted in accordance with established national and international standards (IS 4941 / AOAC / FSSAI). Based on the analytical results obtained, the submitted sample **CONFORMS** to all quality, physico-chemical, sensory, and purity criteria prescribed for natural raw honey. No laboratory evidence of adulteration, exogenous sugar addition, or critical quality failure was detected.

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